

ESG Analytics — Progress Report (Steps 1–6)

Author: Vivek Parashar **Project:** ESG Sustainability Analytics on S&P 500 Companies **Scope of this update:** Data Cleaning → Quality Validation → Feature Engineering → EDA → Risk Index → Next-Year Forecasting **Dataset:** `preprocessed_content.csv` — 866 sustainability-report rows × 10 columns (263 companies, 2014–2023). Source: Kaggle ("ESG Sustainability Reports of S&P 500 Companies"). Text, named entities and E/S/G scores are pre-computed in the dataset; my work builds the full analytics, risk and modelling layer on top of it.

1. Objective

Companies publish 100+ page sustainability reports every year, which are impossible to compare manually. The goal of this project is to turn a pre-scored ESG text dataset into **decision-useful intelligence** — cleaned and validated data, engineered signals, exploratory insights, a company risk-ranking, and a leakage-safe model that forecasts next-year ESG performance. Steps 7–8 (a GenAI copilot and a Streamlit dashboard) are planned next and are not part of this update.

2. Methodology, Reasoning and Results

Each step below states **what** was done, **why**, and the **result**.

Step 1 — Data Cleaning

What: Dropped a stray pandas index column (`Unnamed: 0`); parsed the `ner_entities` string column into real Python lists using `ast.literal_eval` (not `eval`, to avoid arbitrary-code-execution risk); standardised text (`ticker` upper-cased and stripped) and numeric types (`pd.to_numeric(errors="coerce")`); removed duplicate reports keyed on `ticker + year`. **Why:** An index column left in place can leak into the model as a fake feature; string-formatted lists cannot be used as lists; duplicate filings (the same company-year filed on two exchanges) would double-count in every downstream statistic. **Result:** $866 \times 10 \rightarrow 864 \times 9$ (2 duplicate ABT-2021 filings removed). Clean, correctly-typed dataset.

Step 2 — Data Quality Validation

What: Tested the assumed relationship `total_score = e + s + g`, validated score ranges and years, and checked for missing values. **Why:** Assumptions must be verified, not trusted. If `total_score` were exactly the sum of the pillars, using the current pillars to predict the total would be circular (data leakage). **Result (honest finding):** The relationship is **additive and equal-weight** — a linear fit gives coefficients [1.00, 1.00, 1.00] with $R^2 = 0.9993$ — but it is **not an exact identity**. Only 40.0% of rows match to two decimals and 77.5% fall within a rounding-level tolerance (0.02); 99.2% are within 1.0, yet **~22% of rows (194) deviate by more than floating-point rounding can explain (max gap 2.77)**. Conclusion: `total_score` is treated as an *approximately-additive, separately-reported* figure, compared with `np.isclose`/tolerance rather than `==`, with large-gap rows flagged as data-quality outliers. No missing values; all scores non-negative; years within 2014–2023.

Step 3 — Feature Engineering

What: Built **18 derived features** in three groups: - **Text/entity:** `word_count`, `char_count`, `entity_count`, `unique_entities`, `entity_diversity` - **Composition:** `e_share`, `s_share`, `g_share`, `dominant_pillar`, `pillar_std` - **Lag/time:** `prev_total`, `prev_e/s/g`, `roll_mean_total`, `yoy_change`, `years_since_2014`, `report_seq` **Why:** Raw columns are not "model-ready". Text features capture disclosure depth; composition features capture a company's focus (not just size); lag features (built with `shift(1)` inside each company) allow next-year forecasting using **only past information** — the mechanism that keeps the model leakage-free. **Result:** 9 → **27 columns**, saved as `esg_features.csv`.

Step 4 — Exploratory Data Analysis

What: Four targeted charts — pillar score distributions, pillar correlation heatmap, temporal trend (2014–2023), and top-10 companies by average score. **Why:** EDA is about insight, not chart count; each chart answers a specific business question. **Result / insights:** Social is the most-disclosed pillar (mean 10.3) and Environmental the least (mean 5.9); pillars are positively correlated; average disclosure rises over 2014–2023 (regulatory pressure); the top-ranked companies are concentrated in energy/oil & gas (sectors under the most ESG scrutiny). Figures saved in `figures/`.

Step 5 — ESG Risk Index

What: Normalised each pillar to 0–100 (`MinMaxScaler`) and combined them into a weighted index — **Risk Index = $0.40 \cdot E + 0.30 \cdot S + 0.30 \cdot G$** — then bucketed into High/Medium/Low and ranked all companies. **Why:** Executives need a single comparable number and simple bands, not three raw decimals. Environmental is weighted highest because climate is currently the most material ESG risk; the weights are an explicit, configurable business judgment. **Result:** `risk_index` spans 3.6–66.0; **529 Low, 335 Medium, 0 High** (no company crosses the 66 High threshold — an honest artifact of the scale). All **263 companies ranked**; energy names (CTRA, EOG, CVX, PXD) top the list. Saved as `esg_risk.csv`.

Step 6 — Next-Year ESG Forecasting (leakage-safe)

What: Framed a regression problem — predict next year's `total_score` from **past/structural features only** (`prev_total`, `prev_e/s/g`, `roll_mean_total`, `word_count`, `entity_count`, `years_since_2014`, `report_seq`). Explicitly **excluded** current-year `e/s/g/total` and `yoy_change` (since `yoy_change = total - prev_total` hides the target and would give a fake $R^2 = 1.0$). Compared Linear Regression, Random Forest and a naive persistence baseline on an 80/20 split (seed 42). **Why:** A model that only sees the past can actually be deployed; one that peeks at the answer cannot. Beating a naive baseline proves the engineered features carry real signal. **Result (test set, 603 modelling rows):**

Model	MAE	RMSE	R^2
Linear Regression	1.59	2.23	0.913
Random Forest (200)	1.62	2.29	0.908
Naive baseline (last = next)	1.75	2.50	0.891

Both models beat the naive baseline; Linear Regression wins (simplest model generalises best on small, largely-linear data). `prev_total` is by far the strongest predictor. Model saved to `outputs/models/esg_model.pkl`. $R^2 \approx 0.91$ is an **honest** forecasting result, not a leakage-inflated 1.0.

3. Results Summary

Step	Output	Key result
1 Clean	864 × 9 dataset	2 duplicate filings removed, types fixed
2 Validate	Documented assumptions	total $\approx$ e+s+g (coeffs 1,1,1, $R^2=0.999$); <i>approximately</i> additive, not exact
3 Features	27 columns	18 engineered (text / composition / leakage-safe lag)
4 EDA	4 charts	Social highest, Environmental lowest; disclosure rising; energy sector leads
5 Risk Index	<code>esg_risk.csv</code>	263 companies ranked; 40-30-30 weighted index
6 Forecast	<code>esg_model.pkl</code>	Leakage-safe Linear Regression, R^2 0.913, beats baseline

4. Data-Integrity Note

The most important finding of this update is in Step 2: a widely-assumed exact identity (`total = e+s+g`) turned out to be only *approximately* true in the raw data. Rather than hand-wave the gap as "rounding", it is characterised honestly (~22% of rows deviate beyond rounding, up to 2.77) and handled with tolerance-based comparison and outlier flagging. This discipline — verifying assumptions instead of trusting them — underpins the reliability of every later step.

5. Next Steps

- **Step 7 — GenAI Copilot (RAG):** natural-language Q&A grounded on the ESG data.
 - **Step 8 — Streamlit Dashboard:** interactive scores, trends, risk-ranking and the copilot in one app.
 - Optional model upgrades: TF-IDF/FinBERT-based scoring, cross-validation, and sensitivity analysis on the risk weights.
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Reproducibility: all results above are produced end-to-end by `ESG_MethodoLogy_Steps_1-6.ipynb` (runs top-to-bottom with zero errors on the Kaggle dataset).